Q-band ART: Adaptive Band Suppression for Anti-Rollover Tracking of Semi-Trailer Tank Trucks

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Abstract—Semi-trailer tankers are prone to rollover because lateral acceleration excites both suspension roll and liquid sloshing. Existing anti-rollover tracking methods usually introduce liquid states, roll states, or lateral load transfer ratio predictions into model predictive control, which increases dependence on liquid calibration, trailer-side sensing, and online model dimension. This paper proposes Q-band ART (Anti-Rollover Tracking), an adaptive band-suppression framework that penalizes the predicted lateral-acceleration energy in rollover-critical frequency bands. A reduced cart-roll-sloshing model explains why the critical band should be identified from the coupled acceleration-to-risk response, and the resulting Q-band term is embedded into a nonlinear model predictive tracking controller customized for semi-trailer tank trucks with acceleration smoothing, soft band constraints, preview-adaptive weights, extended-state disturbance compensation, hitch-angle-based parameter scheduling. The method is validated in simulations and 1:14 model-truck experiments, reducing peak rollover-risk proxy by xxx% while maintaining acceptable tracking error and real-time performance. Open-source implementation is available at https://github.com/KomasaQi/HitchOpen-THICV-Stack

Index Terms—Anti-rollover control, semi-trailer tanker, liquid sloshing, path tracking, model predictive control, frequency-band suppression.

I. Introduction

SEMI-TRAILER tank trucks are indispensible for liquid hazardous-chemical logistics, but their partially filled tanks make rollover a severe safety risk. Accident reports show that tank truck rollovers are often triggered by sharp steering, emergency avoidance, curved-road overspeed, or driver error, and may lead to deadly secondary disasters [1]–[3]. Because safety standards leave practical operation across partial filling rates [4], [5], free-surface sloshing must be considered in autonomous tank truck tracking.

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Studies show that with free-surface, driving excitations generate additional sloshing, making medium filling rates yielding stronger dynamic instability than either empty or fully filled cases. And the sloshing in tank has been modeled by quasi-static centroid-shift methods, mechanical equivalent models (MEMs), computational fluid dynamics, and experiments. [6]–[10]. Among the methods, MEMs especially pendulum models are widely used for their ability to balance interpretability and real-time computation, although the dominant sloshing modes still require accurate calibration [11]–[14].

For semi-trailer tank trucks, rollover prevention is further complicated by tractor-trailer articulation. High-fidelity models may include lateral, yaw, and roll motion of tractor and trailer,respectively, as well as liquid states [15]–[18]. Such models are useful offline, but online use is limited by trailer replacement, non-unified trailer interfaces, limited trailer-side sensing, and uncertain liquid/-suspension parameters. Active suspension, steering, and differential braking have all been studied for rollover prevention [19]–[21]; tank truck-specific controllers have used PID, LQR, MPC, model-free adaptive control (MFAC), and other robust control with MEMs [22]–[27].

Most existing controllers still rely on explicit liquid states, roll states, or lateral load transfer ratio (LTR) predictions. This is not always suitable for autonomous semi-trailer tank trucks with limited trailer sensing and embedded computing resources. Model predictive control (MPC) can combine tracking, actuator, acceleration, and safety constraints [28], but high-dimensional states and dense constraints challenge real-time implementation [29]. This paper therefore suppresses rollover excitation from the input side: the predicted lateral-acceleration sequence is projected onto vehicle-liquid critical bands, and the resulting energy is penalized and softly constrained. The workflow is shown in Fig. 1.

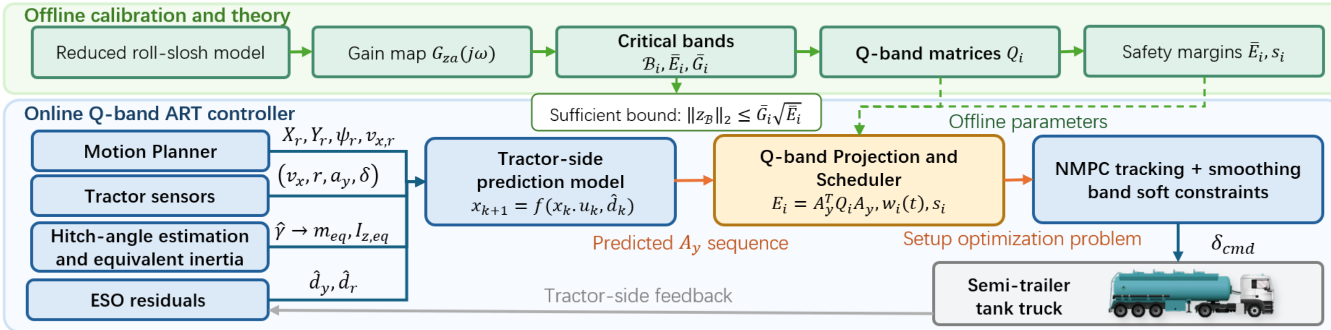

Fig. 1: Architecture of Q-band ART. Offline calibration identifies the critical bands, energy thresholds, Q-band matrices, and conservative margins. The online controller then combines tractor-side prediction, ESO residual compensation, hitch-angle-based equivalent-parameter scheduling, Q-band projection, and NMPC to output steering commands using tractor-side sensing and precomputed Q-band parameters.

The contributions are threefold. First, a Q-band anti-excitation view is introduced for semi-trailer tank truck tracking, avoiding mandatory online liquid-state modeling and observation. Second, an multi-point NMPC tracking controller with disturbance compensation and trailer-free modeling is designed specifically for semi-trailer. Third, the Q-band term is integrated predictively into the controller for anti-rollover purpose and validated through simulation and 1:14 model-truck experiments. The rest of the paper presents the model, controller, validation, and conclusions.

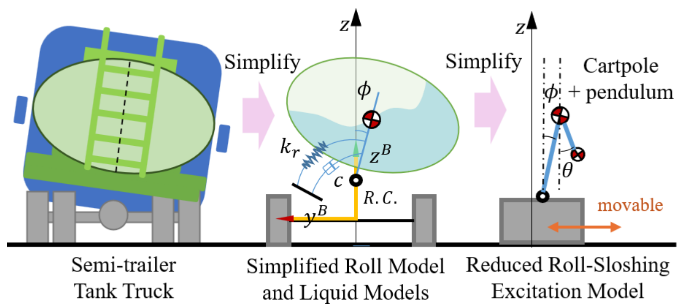

Fig. 2: Reduced lateral model. The online controller only needs the frequency response between input and output.

II. Q-Band Suppression Theory

A. Reduced Roll-Sloshing Excitation Model

Oneline Full liquid state prediction requires observation from liquid [14], which is not always available. To avoid full state prediction, a reduced model is established to explain why only band suppression from input is enough for anti-rollover and how to identify dangerous frequency components. The unsprung lateral motion is represented by a lateral cart, the sprung mass by a roll pendulum, and the dominant liquid mode by a pendulum attached to the tank body. Taken lateral acceleration a_y as input, the coupled roll-sloshing dynamics can be linearized as

$$\mathbf{M}\ddot{\mathbf{x}}_r + \mathbf{C}\dot{\mathbf{x}}_r + \mathbf{K}\mathbf{x}_r = \mathbf{B}_a a_y, \;\; \mathbf{x}_r = [\phi, \theta, \dot{\phi}, \dot{\theta}]^{\mathrm{T}}, \tag{1}$$

where ϕ is the tank-body roll angle and θ is the relative liquid sloshing angle. The off-diagonal terms in $\mathbf{M}$ and $\mathbf{K}$ describe the coupled free mode between suspension roll and liquid motion as

$$\det(\mathbf{K} - \omega^2\mathbf{M}) = 0. \tag{2}$$

Therefore, the dangerous frequency is not the isolated liquid natural frequency or the roll frequency, they appeal to each other and generate a broad-peak.

B. Critical Band and Risk Bound

Let the rollover-risk output be

$$z = C_z\mathbf{x}_r + D_z a_y, \tag{3}$$

where z represents LTR, roll angle, or another rollover indeces, and in this work the roll rate is used as the measurable risk-output. From (1), the input-output frequency response is

$$G_{za}(s) = C_z(s^2\mathbf{M} + s\mathbf{C} + \mathbf{K})^{-1}\mathbf{B}_a + D_z. \tag{4}$$

The Q-band $\mathcal{B}_i = [f_i^-, f_i^+]$ is selected around the broad-peak region of $|G_{za}(j2\pi f)|$. In this work the main band is centered on the dominant sloshing-related response, with additional bands covering secondary and higher-frequency coupled excitations.

Safety could be stated as a sufficient frequency-domain condition. Let $a_{y,\mathcal{B}}$ denote the lateral-acceleration component in a target band $\mathcal{B}$, and let $z_\mathcal{B}$ be the corresponding risk-output component. Around the calibrated operating point, assume that the linearized system is stable and

$$|G_{za}(j2\pi f)| \le \bar{G}_\mathcal{B}, \qquad f \in \mathcal{B}. \tag{5}$$

Then Parseval's relation gives

$$\|z_\mathcal{B}\|_2^2 \le \bar{G}_\mathcal{B}^2 \|a_{y,\mathcal{B}}\|_2^2. \tag{6}$$

Thus, if $\|a_{y,\mathcal{B}}\|_2^2 \le \bar{E}_\mathcal{B}$, where $\bar{E}_\mathcal{B}$ is the given band-energy threshold,

$$\|z_\mathcal{B}\|_2 \le \bar{G}_\mathcal{B}\sqrt{\bar{E}_\mathcal{B}}. \tag{7}$$

Considering static load-transfer contribution and bounded residual terms, a conservative rollover margin can be written as

$$|\mathrm{LTR}_{\mathrm{stat}}| + \bar{G}_{\mathcal{B}}\sqrt{\bar{E}_{\mathcal{B}}} + R_\perp + R_0 + R_m \le \mathrm{LTR}_{\mathrm{lim}}, \quad (8)$$

where $R_\perp$, R_0, and R_m bound non-target-band excitation, initial roll/slosh energy, and model or sensing errors, respectively. Proof: Inequality (6) follows from the frequency-domain gain bound of the stable linearized input-output model. Substituting the band-energy threshold gives (7). Adding the bounded static, off-band, initial-condition, and model-error contributions yields (8). The condition is sufficient rather than necessary, and its conservatism comes from linearization, band discretization, gain upper bounding, and residual margins.

This proof explains why limiting the band energy of a_y can reduce rollover-risk excitation even when liquid phase and true wheel loads are not observed online. The discrete Q-band matrix used by the controller is an implementable approximation of $\|a_{y,\mathcal{B}}\|_2^2$ over the prediction horizon.

III. Adaptive Q-Band Suppression Controller

A. Tractor Prediction Model and ESO

The online optimizer uses a low-dimensional tractor-side nonlinear prediction model and treats trailer force, liquid sloshing, tire mismatch, and parameter error as lumped residuals. For the dynamic version, the front and rear tire slip angles are

$$\alpha_f = \delta - \frac{v_y + l_f r}{v_x}, \qquad \alpha_r = -\frac{v_y - l_r r}{v_x}, \quad (9)$$

$$F_{yf} = C_f \alpha_f, \qquad F_{yr} = C_r \alpha_r, \quad (10)$$

where v_x, v_y, r, and δ are longitudinal speed, lateral speed, yaw rate, and front steering angle. The nominal prediction model is

$$\begin{aligned} \dot{v}_y &= \frac{F_{yf}\cos\delta + F_{yr}}{m_{\mathrm{eq}}} - v_x r + d_y, \\ \dot{r} &= \frac{l_f F_{yf}\cos\delta - l_r F_{yr}}{I_{z,\mathrm{eq}}} + d_r, \end{aligned} \quad (11)$$

The extended-state observer estimates the slow-varying residuals. An extended state observer (ESO) is used in yaw channel:

$$\begin{aligned} \dot{\hat{r}} &= \frac{l_f \hat{F}_{yf}\cos\delta - l_r \hat{F}_{yr}}{I_{z,\mathrm{eq}}} + \hat{d}_r + \beta_{r1}(r - \hat{r}), \\ \dot{\hat{d}}_r &= \beta_{r2}(r - \hat{r}). \end{aligned} \quad (12)$$

The lateral distrubance mainly consist of lateral slope and could be compensated as the residual of measured lateral acceleration and centrifugal acceleration

$$\hat{d}_y = a_y^{centri} - a_y^{meas} = v_x r - (a_y - \Delta a_{y,0}), \quad (13)$$

where $\Delta a_{y,0}$ is the zero-offset of IMU and could be adjusted slowly according to constant bias of real tracking error after compensation. In prediction, the estimated residual could be held or exponentially decayed,

$$\hat{d}_{\ell|k} = \lambda_d^{\ell}\hat{d}_k, \qquad 0 < \lambda_d \le 1, \quad (14)$$

so that recent unmodeled trailer and liquid effects improve short-horizon tracking without forcing the optimizer to identify liquid states.

B. Hitch-Angle and Equivalent Inertia

The equivalent parameters in (11) require a tractor-trailer articulation prior. Let the tractor and trailer headings be ψ_1 and ψ_2, and define the hitch angle as

$$\gamma = \psi_1 - \psi_2. \quad (15)$$

With the tractor yaw rate measured as r_m, the trailer heading and hitch angle are estimated by

$$\begin{aligned} \dot{\hat{\psi}}_2 &= K_v \frac{v_x}{L_2}\left(\frac{L_h}{L_1}\tan\delta\cos\hat{\gamma} + \sin\hat{\gamma}\right), \\ \dot{\hat{\gamma}} &= r_m - \dot{\hat{\psi}}_2. \end{aligned} \quad (16)$$

Here L_1 is the tractor wheelbase, L_2 is the distance from the hitch to the trailer axle group, and L_h is the distance from tractor rear axle to hitch. The correction

$$K_v = 1 - \frac{m_2 v_x^2 e}{L_1 L_2 k_3} \quad (17)$$

keeps a first-order effect of trailer-axle sideslip; $K_v = 1$ is used when the trailer tire parameters are unavailable.

The trailer contribution is then projected into tractor lateral inertia and yaw inertia:

$$m_{x,\mathrm{eq}}(\hat{\gamma}) = m_1 + m_2\cos^2\hat{\gamma}, \qquad m_{y,\mathrm{eq}}(\hat{\gamma}) = m_1 + m_2\sin^2\hat{\gamma}, \quad (18)$$

$$I_{z,\mathrm{eq}}(\hat{\gamma}) = I_{z1} + I_{z2}\left(\frac{d_1}{d_2}\right)^2\cos^2\hat{\gamma} + m_2 d_1^2\sin^2\hat{\gamma}. \quad (19)$$

In the NMPC model, $m_{\mathrm{eq}} = m_{y,\mathrm{eq}}(\hat{\gamma})$, and $I_{z,\mathrm{eq}}$ is evaluated from (19), after low-pass filtering and saturation. When $\hat{\gamma} \approx 0$, the trailer mainly follows the tractor longitudinally and contributes less instantaneous lateral inertia; as the articulation grows, more trailer inertia is projected into the tractor lateral/yaw response. This is a scheduling prior rather than a full articulated-vehicle model, and the remaining fast coupling is handled by the ESO and Q-band safety margins.

C. Band-Energy Projection

Let the prediction horizon contain N samples and sampling time Δt. The predicted lateral-acceleration sequence is

$$\mathbf{A}_y = [a_{y,0}, a_{y,1}, \ldots, a_{y,N-1}]^{\mathrm{T}}. \quad (20)$$

For a critical band $\mathcal{B}_i$, choose discrete frequencies f_{ij} and time grid $t_k = k\Delta t$. Define

$$\mathbf{c}_{ij} = [\cos(2\pi f_{ij} t_0), \ldots, \cos(2\pi f_{ij} t_{N-1})]^{\mathrm{T}}, \quad (21)$$

$$\mathbf{s}_{ij} = [\sin(2\pi f_{ij} t_0), \ldots, \sin(2\pi f_{ij} t_{N-1})]^{\mathrm{T}}. \quad (22)$$

The band energy is

$$E_i(\mathbf{A}_y) = \mathbf{A}_y^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_y, \tag{23}$$

$$\mathbf{Q}_i = \frac{1}{N^2} \sum_j (\mathbf{c}_{ij}\mathbf{c}_{ij}^{\mathrm{T}} + \mathbf{s}_{ij}\mathbf{s}_{ij}^{\mathrm{T}}) \succeq 0. \tag{24}$$

The matrix $\mathbf{Q}_i$ can be precomputed for fixed Δt, N, and $\mathcal{B}_i$. Hence the Q-band term adds no new vehicle state; it only reshapes the predicted lateral acceleration.

D. Optimization Problem

The Q-band ART objective combines tracking, actuator regularization, acceleration smoothing, and band suppression:

$$\begin{aligned} \min_{\mathbf{U},\mathbf{S}} \; & J_{\mathrm{trk}} + J_u + J_{\Delta u} + w_a \|\mathbf{A}_y\|_2^2 + w_1 \|\mathbf{D}_1 \mathbf{A}_y\|_2^2 \\ & + w_2 \|\mathbf{D}_2 \mathbf{A}_y\|_2^2 + \sum_i w_i(t) \mathbf{A}_y^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_y + \sum_i \rho_i s_i^2. \end{aligned} \tag{25}$$

where $\mathbf{D}_1$ and $\mathbf{D}_2$ are first- and second-order difference matrices, J_{trk} is the point-wise euclidean tracking error and yaw error

$$J_{\mathrm{trk}} = w_{xy}(\|\mathbf{X} - \mathbf{X}_{\mathrm{ref}}\|_2^2 + \|\mathbf{Y} - \mathbf{Y}_{\mathrm{ref}}\|_2^2) + w_\psi(\|\psi - \psi_{\mathrm{ref}}\|_2^2). \tag{26}$$

The constraints are

$$\begin{aligned} & \mathbf{x}_{k+1} = f(\mathbf{x}_k, u_k, \hat{d}_{k|k}, m_{\mathrm{eq}}, I_{z,\mathrm{eq}}), \\ & |a_{y,k}| \le a_{y,\max}^{\mathrm{dyn}}, \\ & u_{\min} \le u_k \le u_{\max}, \quad |\Delta u_k| \le \Delta u_{\max}, \\ & E_i(\mathbf{A}_y) \le \bar{E}_i + s_i, \quad s_i \ge 0. \end{aligned} \tag{27}$$

The soft band constraint prevents the optimizer from concentrating energy in a dangerous band, while preserving feasibility when the reference path is aggressive. In low-risk segments, $w_i(t)$ remains small and the controller behaves like a normal tracker.

E. Preview-Adaptive Scheduling

Pure feedback scheduling raises the safety weight only after the roll-slosh response has already been excited. Q-band ART uses preview risk as the main trigger. A reference lateral-acceleration sequence $\mathbf{A}_{y,\mathrm{ref}}$ is computed from the future path, and its band energy is compared with the calibrated threshold:

$$R_i^{\mathrm{pre}} = \max\left(0, \frac{\mathbf{A}_{y,\mathrm{ref}}^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_{y,\mathrm{ref}}}{\bar{E}_{i,\mathrm{ref}}} - 1\right). \tag{28}$$

The raw weight is scheduled as

$$w_i^{\mathrm{raw}} = w_i^{\min} + \left(1 - e^{-\kappa R_i^{\mathrm{pre}}}\right)(w_i^{\mathrm{tar}} - w_i^{\min}). \tag{29}$$

Observed yaw, acceleration, or roll residuals can further increase the weight when residual sloshing remains. A non-symmetric filter is then applied:

$$w_i(t) = (1 - \alpha_i) w_i(t - \Delta t) + \alpha_i w_i^{\mathrm{raw}}, \tag{30}$$

where $\alpha_i = \alpha_\uparrow$ if $w_i^{\mathrm{raw}} > w_i(t - \Delta t)$ and $\alpha_i = \alpha_\downarrow$ otherwise. With $\alpha_\uparrow > \alpha_\downarrow$, the weight rises before a high-risk lane change and decays slowly after it.

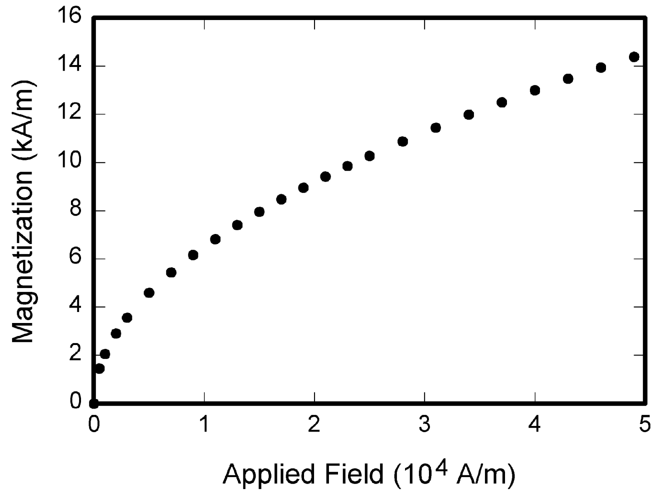

Fig. 3: Adaptive Q-band scheduling. Preview risk increases band weights before aggressive steering, while residual monitoring delays weight release after the maneuver.

F. Implementation Notes

The Q-band module is model-agnostic. For a kinematic predictor, $a_{y,k}$ can be approximated from curvature or steering; for a dynamic predictor, it can be computed from $\dot{v}_y + v_x r$; for TruckSim or model-truck validation, it is evaluated from the predicted tracking response. In all cases, the controller suppresses the lateral excitation sequence rather than explicitly controlling the liquid state.

IV. Validation

A. 5-Pins Simulation Setup

The proposed controller was first evaluated in a 5-pins lane-change simulation. The reference path contains five consecutive double-lane-change maneuvers with the same lateral displacement of 3.5 m and progressively shorter maneuver durations, starting from 4 s. This setting produces a continuous risk gradient: the first two double lane changes are mild, the third is transitional, and the fourth and fifth contain strong lateral excitation. The controlled object follows the semi-trailer tank truck setting used in TruckSim validation, while the Q-band module only uses the predicted lateral-acceleration sequence in the controller.

Fifteen controller variants were compared, including baseline tracking, lateral-acceleration hard constraint, acceleration magnitude/difference penalties, fixed Q-band, preview-adaptive Q-band, feedback-assisted Q-band, and Q-band soft constraints. The main metrics are peak $|\mathrm{LTR}_{\mathrm{est}}|$, RMS LTR, peak lateral acceleration, RMS tracking error, and segment-wise tracking/safety indicators.

B. Simulation Results

Fig. 4 shows that all controllers track the first three maneuvers with small errors. As the maneuver duration

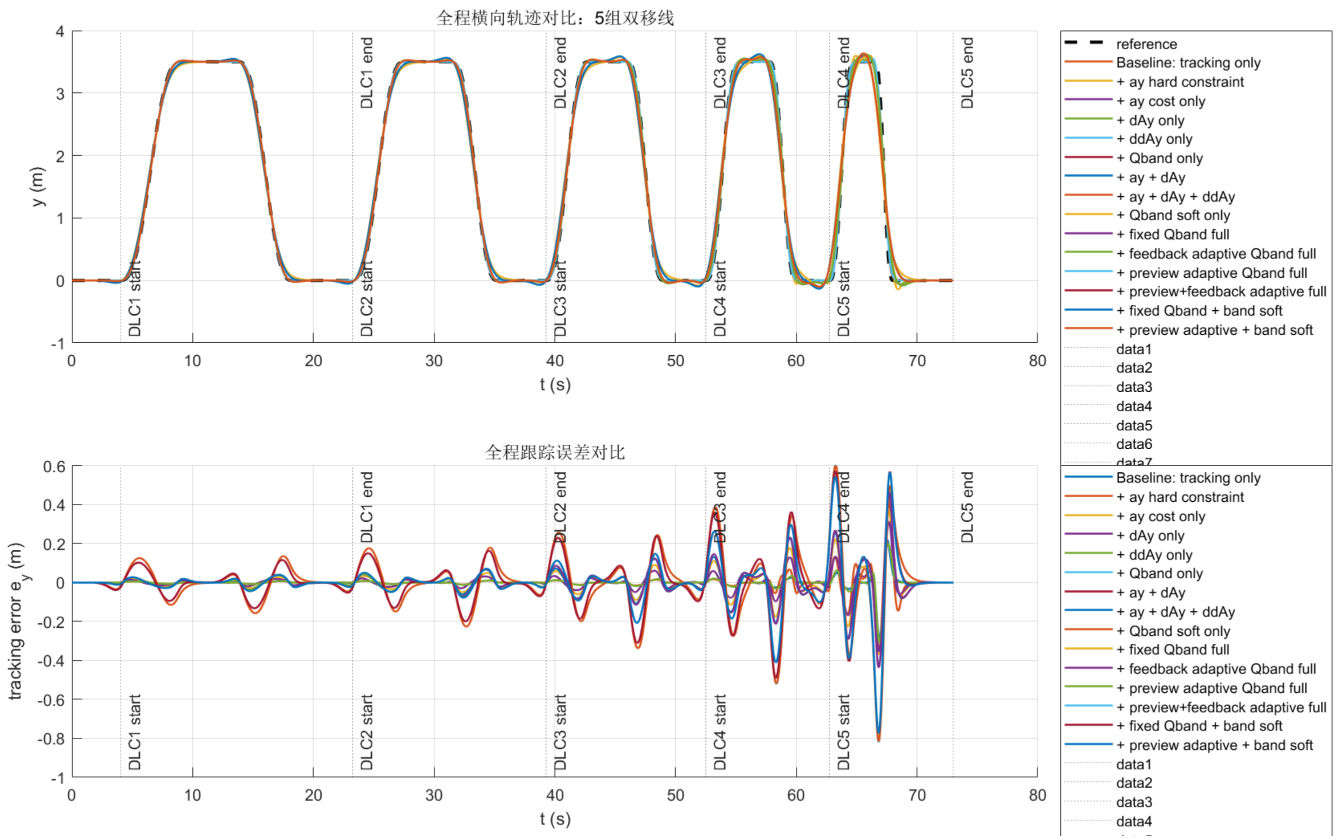

Fig. 4: Five-pins tracking results. Low-risk segments are tracked closely by all methods; in the high-risk DLC4–DLC5 segments, Q-band and soft-constraint variants intentionally allow moderate path deviation to reduce rollover excitation.

decreases, the baseline tracker maintains path accuracy but injects stronger oscillatory lateral acceleration. Q-band methods begin to relax tracking in DLC4 and DLC5, especially near the lane-return phases, where the predicted acceleration would otherwise fall into the critical band. The largest temporary tracking deviation is approximately xxx m in the current simulation and remains acceptable for the intended emergency maneuver envelope.

The ablation results in Fig. 5 show three trends. First, the baseline peak $|\mathrm{LTR}_{\mathrm{est}}|$ reaches about xxx, exceeding the rollover-risk limit. Second, acceleration magnitude and difference penalties reduce part of the risk but cannot consistently suppress the critical high-risk segments. Third, fixed or adaptive Q-band variants reduce peak $|\mathrm{LTR}_{\mathrm{est}}|$ to approximately xxx–xxx and lower RMS LTR from xxx to xxx. The preview-adaptive and soft-constraint variants provide the most consistent reduction because the weights rise before DLC4–DLC5 and the optimizer is discouraged from placing energy in the critical band.

Segment-wise results in Fig. 6 further confirm that Q-band ART is mainly activated in the dangerous sections. In DLC1 and DLC2, all methods have similar peak $|\mathrm{LTR}|$ and tracking error. In DLC3 the Q-band methods begin to separate from acceleration-only regularization. In DLC4 and DLC5, the baseline and acceleration-only strategies approach or exceed the risk threshold, while Q-band variants keep the peak risk proxy near xxx. This supports the design goal: low-risk segments remain close to ordinary tracking, and high-risk segments are reshaped to suppress dangerous lateral excitation.

C. Model-Truck Experiment

The 1:14 semi-trailer tank truck experiment was conducted to verify that the Q-band mechanism remains effective on a physical platform. The experiment used a double-lane-change or 5-pins-like reference path scaled according to the model-truck operating envelope. Baseline tracking and Q-band ART were compared under the same speed and filling-rate settings. The measured responses include path tracking, steering command, lateral acceleration, roll/yaw response, liquid-surface motion when available, and controller solve time.

The model-truck results show the same qualitative behavior as the simulation. Baseline tracking follows the reference more tightly but excites larger roll-slosh motion, and the anti-roll support/contact or rollover-risk proxy reaches xxx. Q-band ART reduces the peak risk proxy by xxx%, keeps the liquid/roll response within xxx, and increases RMS tracking error from xxx m to xxx m. The controller average and peak solve times are xxx ms and

Fig. 5: Ablation summary for the 5-pins simulation. Q-band-based variants reduce both peak $|\mathrm{LTR}_{\mathrm{est}}|$ and RMS LTR compared with baseline tracking, while the safety improvement is obtained at the cost of controlled tracking relaxation.

TABLE I: Validation Summary With Replaceable Results

Case	Peak risk proxy	RMS error	Solve time
Simulation baseline	xxx	xxx m	xxx ms
Simulation Q-band ART	xxx	xxx m	xxx ms
Model-truck baseline	xxx	xxx m	–
Model-truck Q-band ART	xxx	xxx m	xxx ms

xxx ms, respectively, satisfying the real-time requirement of the experimental platform.

V. Conclusion

This paper proposed Q-band ART, an adaptive frequency-band suppression method for anti-rollover tracking of semi-trailer tank trucks. Instead of requiring liquid sloshing states or wheel-load-based LTR as online controller states, the method suppresses the predicted lateral-acceleration energy in critical vehicle-liquid bands. A reduced roll-sloshing model provides the frequency-domain safety interpretation, and the Q-band term is integrated with acceleration smoothing, soft constraints, preview-adaptive scheduling, ESO disturbance compensation, and hitch-angle-based parameter scheduling.

The 5-pins simulation and 1:14 model-truck experiment show that Q-band ART reduces peak rollover-risk proxy from xxx to xxx and improves RMS LTR/risk response by xxx%, while keeping tracking error and computation time within acceptable ranges. The results indicate that input-side critical-band suppression is a practical complement to explicit rollover-state constraints, especially when trailer-side sensing and liquid-state observation are limited. Future work will refine critical-band calibration across filling rates and integrate Q-band scheduling with combined steering and braking control.

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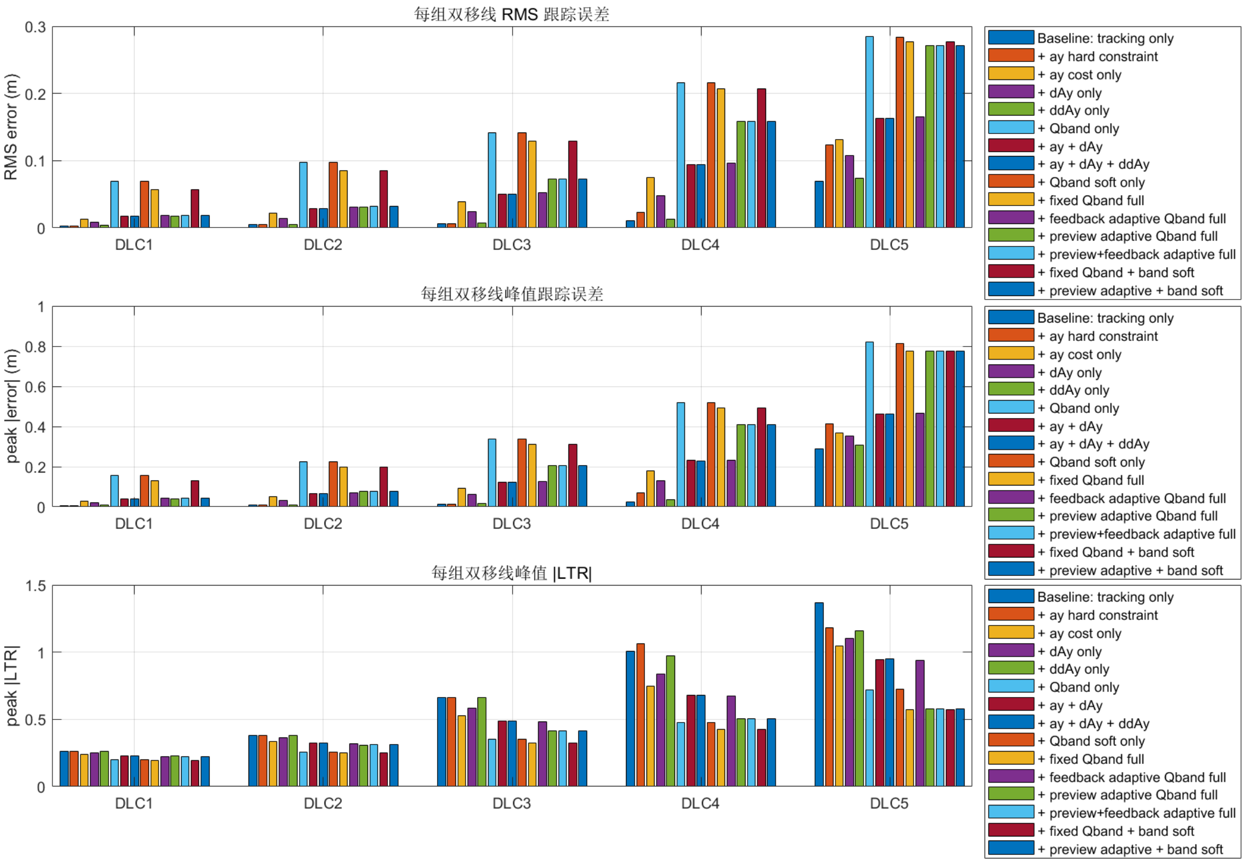

Fig. 6: Segment-wise 5-pins metrics. The difference among methods is small in DLC1–DLC2, becomes visible in DLC3, and is dominated by the safety-tracking tradeoff in DLC4–DLC5.

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